

5.4 Writing and Solving Real-World Equations

Lesson Introduction

 Benchmark	MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.		 Learning Target	I can write and solve two-step equations given a real-world context.
 Benchmark Coherence	Building On: MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context where all terms are rational numbers.		Working Toward: N/A	
 Essential Question	How can I write and solve two-step equations given a real-world context?			
 Lesson Narrative	In this lesson, students will extend their learning from the prior lesson to write and solve two-step equations given a real-world context. Students will be introduced to a scaffolded instructional approach to solving real-world problems by writing and solving equations. They will get the opportunity to practice with multiple real-world scenarios.			
 Component Summary	5.4.1 Warm-Up (~5 Min)	Hanger diagrams activity (<i>SE p. 244</i>)	5.4.3 Guided Practice (10-15 min)	Practice solving real-world problems by writing and solving equations (<i>SE p. 246</i>)
	5.4.2 Guided Instruction (10-15 min)	Introduction to solving real-world problems by writing and solving equations (<i>SE p. 245</i>)	5.4.4 Try It (10-15 min)	Practice solving real-world problems by writing and solving equations (<i>SE p. 247</i>)
	5.4.5 Wrap-Up (~5 min)	Solving real-world problems by writing and solving equations (<i>SE p. 248</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may believe time should always be represented on the x-axis or always the y-axis. - Students may believe only one parameter can exist per situation, like questions that can only have one answer.	- Students will engage in real-world problems throughout this entire lesson. Consider leveraging your existing word problem strategies throughout the lesson to support students as they engage in word problems. - Watch the pacing of this lesson as it consists solely of word problems. Consider the use of a timer to maintain lesson momentum.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson provides students the opportunity to collaborate on real-world problems. For this lesson, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.	MA.K12.MTR.7.1: Real-World Contexts This lesson consists solely of real-world problems. During this lesson, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.
 Differentiate Instruction	Hanger diagrams, as pictured in the warm-up, can be powerful tools for conceptualizing the process of solving equations with variables. For students in need of additional support with solving equations, consider a station for students to visit during 5.4.4 Try It! that extends the warm-up, providing students digital manipulatives to explore hanger diagrams for additional scaffolding in solving equations. This station could also feature On-Ramp to Algebra 1 Study Expert videos on solving equations for even more support.	
Lesson Notes:		

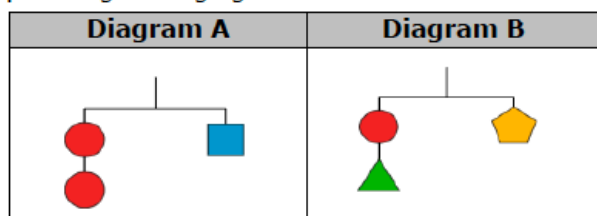
5.4.1 Warm-Up: Hanger Diagrams

Component Overview: Students will complete a hanger diagram puzzle to maintain balance using the information provided in Diagrams A and B.



5.4.1 Warm-Up: Hanger Diagrams

Two hanger diagrams are shown. They are balanced with an equal weight hanging from both sides.



1. Use the diagrams above to complete the right side of each diagram so that Diagrams C and D are balanced. You cannot copy the left side of the diagram.

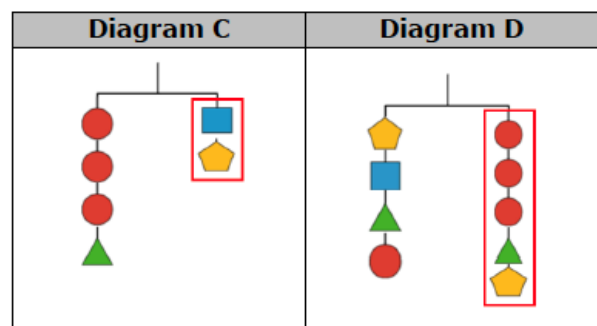


Diagram C: 1 square and 1 pentagon

Diagram D: 3 circles, 1 pentagon, and 1 triangle

2. James drew a pentagon and a square for the right side of Diagram C. Draw pictures or explain how James could conclude that a pentagon and a square are equivalent to the left side of Diagram C.



The weight of two circles is equal to the weight of one square. The weight of a circle and a triangle is equal to the weight of a pentagon.



Consider leveraging a Think-Pair-Share instructional routine with this activity. Allow students individual think time to try and determine their own method for finding an answer. Then allow them time to work with a partner to share their methods and answers.

5.4.2 Guided Instruction: Solving Real-World Problems by Writing and Solving Equations

Component Overview: Students will be introduced to solving real-world problems by writing and solving equations using two real-world scenarios.



5.4.2 Guided Instruction: Solving Real-World Problems by Writing and Solving Equations

1. The U.S. Census Bureau recorded a population of approximately 18.8 million people living in Florida in 2010. Since 2010, there has been an average increase of 0.3 million people per year.
 - a. What are the quantities that are changing in this scenario?
People, in millions, per year.
 - b. Define the variables a and p .
Answers will vary. a represents the number of years since 2010. p represents the number of people living in Florida, in millions.
 - c. What is the parameter?
0.3
 - d. What is the constant?
18.8
 - e. Which equation represents the situation?
Answers will vary.
 - $p = 18.8a + 0.3$
 - $p = 0.3a + 18.8$
 - $a = 18.8p + 0.3$
 - $a = 0.3p + 18.8$
 - f. Use your equation to determine in what year Florida's population was approximately 21.2 million.
2018



The correct equation for question 1e will depend on how the variables are defined in part b.

If a is defined as “years since 2010,” and p as the “number of people living in Florida” in millions, then the second equation is correct.

If the variables are reversed, then the 4th equation is correct.



Consider facilitating a dialogue with students about the multiple correct answers in question 1e to reinforce the importance of defining variables when writing equations to represent a real-world context.

5.4.2 Guided Instruction: Solving Real-World Problems by Writing and Solving Equations (continued)

2. Simone purchased 4 skinny vanilla lattes for her and her three friends. A 6% sales tax was applied to the purchase, and she gave the rest of her money as a \$3.86 tip for a total of \$24.

- a. Define the variable in this context.

The price per latte

- b. Discuss with your partner what the parameters are in this context.

- c. Complete the table to write an equation that represents the context.

Total \$ Spent	=	# of Lattes	(Latte Price)	(1 + Tax Rate)	+	Tip
24	=	4	(x)	(1 + 0.06)	+	3.86

- d. Solve the equation for the unknown.

4.75

- e. Interpret the solution.

Each latte costs \$4.75.



Turn and Talk

Prompt a brief Turn and Talk connecting price per latte and taxes to parameters. Students may only identify the price per latte as a parameter. Monitor for students with different answers and use talk moves to facilitate discussion.



How did you determine your interpretation of the solution?

5.4.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

Component Overview: Students will practice solving a real-world problem by writing and solving equations using one real-world problem.



5.4.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

1. The popularity of Twitter can be measured using the number of active accounts. In January of 2015, Twitter had approximately 302.5 million active accounts worldwide. Each year, the number of active accounts has increased by about 10.2 million.
 - a. Define the variables in this context.
 $a = \text{total active accounts}$
 $n = \text{number of years since 2015}$
 - b. What is the parameter?
 10.2
 - c. What is the constant?
 302.5
 - d. Which variable is the output in this context?
 a
 - e. Write an equation to represent this situation.
 $a = 10.2n + 302.5$
 - f. Write a one-variable equation that can be used to determine the year Twitter will have 425 million active accounts if the trend continues.
 $425 = 10.2n + 302.5$
 - g. Solve the equation from Part F.
 12
 - h. Interpret the solution.
In 2027, Twitter will have 425 million active users if this trend continues.
 - i. Are there any constraints to the given situation?
 $n \geq 0$
 $a \geq 0$



For this activity, consider releasing students to work with a partner to answer the scaffolded questions of this activity. While they are working, circulate the room to ensure they are on the right track as they are answering.



Why does this constraint exist?

5.4.4 Try It: Solving Real-World Problems by Writing and Solving Equations

Component Overview: Students will try solving a real-world problem by writing and solving equations.



5.4.4 Try It: Solving Real-World Problems by Writing and Solving Equations

1. Since the fourth quarter of 2014, the flash storage capacity of smartphones has grown by approximately 12 gigabytes (GB) a year. The average flash storage in 2014 was 15.9 GB.

- a. Write an equation to represent this situation, defining both variables.

$s = \text{total storage}$

$y = \text{years since 2014}$

$S = 12y + 15.9$

- b. Assuming the trend continues, write a one-variable equation that can be used to find the average flash storage capacity of smartphones in 2025.

$S = 12(11) + 15.9$

- c. Solve your equation from Part B.

147.9 GB

- d. Interpret the solution.

The total flash capacity in 2025 will be 147.9 gigabytes.

- e. Discuss any constraints of the situation with your partner.

$y \geq 0$ and $s \geq 0$



For this activity, consider having students try this on their own from start to finish. Then have them compare their work with a partner to discuss their methods and answers to come to a consensus on each of their answers.



- Why does this constraint exist in this scenario?
- Could the situation be modified to make a negative value viable in this context?



Prompt students to consider what constraints might exist relating to the number of years.

5.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a ticket out the door on solving a real-world problem using equations. Students should be allowed to access a calculator to complete the 5.4.5 Wrap-Up.



5.4.5 Wrap-Up: Ticket Out the Door

1. James is going to play paintball. He is charged \$25.95 for equipment rental, plus \$6.57 per box of 100 paintballs. If James spent \$58.80, how many boxes of paintballs did he purchase?

- a. Write and solve an equation representing this situation.

$$58.80 = 6.57x + 25.95$$

$$x = 5$$

- b. Interpret the solution.

James purchased 5 boxes of paintballs.



MA.K12.MTR.2.1: Multiple Representations

Encourage creativity in how students are interpreting the word problem, prompting students to model their thinking. Students sometimes make better sense of context by drawing, sketching, or modeling first, followed by transitioning to the abstract representation of an equation.